$$\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2} \tag{3.10}$$

$$\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2} \tag{3.11}$$

$$\cos C - \cos D = 2\sin\frac{C+D}{2}\sin\frac{D-C}{2} \tag{3.12}$$

Identity 3.15: Prove that $\sin(60^\circ - A)\sin A\sin(60^\circ + A) = \frac{1}{4}\sin 3A$

Proof.

$$\begin{aligned}\sin(60^\circ - A)\sin A\sin(60^\circ + A) &= \sin(60^\circ - A)\sin(60^\circ + A)\sin A \\ &= \frac{1}{2}\left[\cos 2A - \cos 120^\circ\right]\sin A \\ &= \frac{1}{2}\left[\cos 2A\sin A + \frac{1}{2}\sin A\right] \\ &= \frac{1}{2}\left[\frac{1}{2}\sin 3A\right] = \frac{1}{4}\sin 3A\end{aligned}$$

Similarly we can prove the following two important identities

Identity 3.16: $\cos(60^\circ - A)\cos A\cos(60^\circ + A) = \frac{1}{4}\cos 3A$

Identity 3.17: $\tan(60^\circ - A)\tan A\tan(60^\circ + A) = \tan 3A$

Example 3.33 Express each of the following product as a sum or difference
(i) $\sin 40^\circ\cos 30^\circ$ (ii) $\cos 110^\circ\sin 55^\circ$ (iii) $\sin\frac{x}{2}\cos\frac{3x}{2}$

Solution:

(i) We know that $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

Take $A = 40^\circ$and $B = 30^\circ$. We get,

$2\sin 40^\circ\cos 30^\circ = \sin(40^\circ + 30^\circ) + \sin(40^\circ - 30^\circ) = \sin 70^\circ + \sin 10^\circ$.

Thus, $\sin 40^\circ\cos 30^\circ = \frac{1}{2}\left[\sin 70^\circ + \sin 10^\circ\right]$.

(ii) We know that $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

Take $A = 110^\circ$ and $B = 55^\circ$. We get,

$2\cos 110^\circ\sin 55^\circ = \sin(110^\circ + 55^\circ) - \sin(110^\circ - 55^\circ)$.

Thus, $\cos 110^\circ\sin 55^\circ = \frac{1}{2}\left[\sin 165^\circ - \sin 55^\circ\right]$

(iii) We know that $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

Take $A = \frac{x}{2}$ and $B = \frac{3x}{2}$

We get, $2\sin\frac{x}{2}\cos\frac{3x}{2} = \sin\left(\frac{x}{2} + \frac{3x}{2}\right) + \sin\left(\frac{x}{2} - \frac{3x}{2}\right)$

Thus, $\sin\frac{x}{2}\cos\frac{3x}{2} = \frac{1}{2}\left[\sin 2x - \sin x\right]$.